

Dynamic ETL Automation Platform - Business Requirements

1. Business need

The business receives ZIP packages containing multiple client Excel workbooks. Each workbook can contain Empirical Model sheets, Regression Model sheets and a Datasheet. Quarterly model values must be consolidated into a standardized output workbook and persisted into relational databases.

The process must be automated, secure, repeatable and operationally supportable.

2. Goals

- automate the complete ingest -> extract -> transform -> validate -> publish -> archive flow;
- preserve integrity and atomicity of output/database writes;
- support any reasonable number of source files/supported sheets within a batch;
- complete a normal batch in ≤ 30 minutes;
- support safe same-day reruns;
- retain traceability from source package to output rows;
- expose failures with actionable diagnostics;
- deploy through CI/CD and repeatable infrastructure/configuration.

3. Scope

In scope

- secure file ingestion;
- ZIP validation/extraction;
- Excel parsing;
- Empirical/Regression correlation by type;
- output Excel generation;
- relational persistence;
- Datasheet persistence;
- archive/quarantine/cleanup;
- scheduling/event triggering;
- CI/CD;
- operational telemetry;
- troubleshooting/runbook.

Out of scope by default

- redesigning the business calculation formulas inside the source workbooks;
- editing source workbooks;
- human identity/access-management platform beyond project needs;
- graphical ETL designer;
- production customer data in the public training repository.

4. Secure ingestion requirements

ID	Requirement	Priority	Legacy mapping
DETL-ING-001	Provide an automated, documented ingestion endpoint/location on an approved cloud or lab platform.	Must	BR1

ID	Requirement	Priority	Legacy mapping
DETL-ING-002	Use encrypted transport such as SFTP/SSH or object-storage HTTPS. Plain FTP is legacy compatibility only and is not the reference design.	Must	BR1 modernized
DETL-ING-003	Protect persistent source/output storage with encryption at rest where the platform supports it; a Linux block-device implementation may use encrypted volume/LUKS.	Must	BR2
DETL-ING-004	Validate file type, ZIP readability, size limits and expected contents before processing.	Must	Gap filled
DETL-ING-005	Generate a SHA-256 checksum and run manifest for each accepted source archive.	Must	Gap filled
DETL-ING-006	Process each run in an isolated working directory.	Must	BR3/BR6 clarified
DETL-ING-007	Move invalid/unprocessable inputs to quarantine with a machine-readable failure reason.	Must	Gap filled

5. Orchestration requirements

ID	Requirement	Priority	Legacy mapping
DETL-ORCH-001	Support scheduled execution using a systemd timer/cron or equivalent cloud scheduler.	Must	BR3
DETL-ORCH-002	Trigger an orchestration endpoint that validates the batch, extracts the ZIP and invokes the Python ETL process.	Must	BR3
DETL-ORCH-003	Assign every execution a unique <code>run_id</code> .	Must	Gap filled
DETL-ORCH-004	Prevent the same source archive from being processed concurrently unless explicitly configured.	Must	Gap filled
DETL-ORCH-005	Use explicit terminal states such as <code>completed</code> , <code>failed</code> , <code>quarantined</code> and <code>partial</code> if partial mode is allowed.	Must	Gap filled

6. Transformation requirements

ID	Requirement	Priority	Legacy mapping
DETL-TR-001	Extract Excel files from the accepted ZIP into the run working directory.	Must	BR3
DETL-TR-002	Detect all supported Empirical Model and Regression Model sheets in each workbook.	Must	BR9-BR12
DETL-TR-003	Determine the Type represented by each model sheet from the sheet naming/business rule.	Must	BR11, BR15-BR17
DETL-TR-004	When no type is present, store Type as NULL.	Must	BR17
DETL-TR-005	Correlate Empirical and Regression data for the same type into one output record.	Must	BR26
DETL-TR-006	Extract Quarter and Year from the Regression Model according to the validated workbook schema.	Must	BR18-BR19
DETL-TR-007	Extract Estimated Total Sold, Estimated Sold Max and Estimated Sold Min from the Empirical Model.	Must	BR9, BR20-BR22
DETL-TR-008	Extract Forecast w/o SA Actual, Forecast w/o SA Max and Forecast w/o SA Min from the Regression Model.	Must	BR10, BR23-BR25
DETL-TR-009	Derive Ticker /company code from the source filename using a documented parsing rule.	Must	BR13
DETL-TR-010	Emit a separate output record for each source workbook + type combination.	Must	BR14-BR15
DETL-TR-011	Parse the Datasheet and extract the latest month's records into the defined Data dataset.	Must	BR29
DETL-TR-012	Prefer semantic labels/anchors in the workbook over naked hard-coded coordinates; if cell coordinates are required, centralize them in versioned schema configuration.	Must	BR18-BR25 clarified

DETL-TR-013	Detect unsupported/missing/duplicated model pairs and fail or quarantine according to a documented rule rather than silently producing incomplete data.	Must	Gap filled
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7. Output workbook requirements

ID	Requirement	Priority	Legacy mapping
DETL-OUT-001	Write the consolidated output under the configured data-output location.	Must	BR4
DETL-OUT-002	Preserve the canonical filename <code>model_YYYYMMDD.xlsx</code> for the current run date.	Must	BR5
DETL-OUT-003	Resolve the legacy same-day filename collision by writing atomically and storing each historical run under a unique run-ID/timestamp archive path.	Must	BR8 + BR28 clarified
DETL-OUT-004	Output columns must match the approved data dictionary and ordering.	Must	BR13
DETL-OUT-005	Do not publish the final canonical workbook until all mandatory validations pass.	Must	Reliability NFR clarified

8. Database requirements

ID	Requirement	Priority	Legacy mapping
DETL-DB-001	Persist consolidated model output to a relational dataset logically named AnalystData .	Must	BR27
DETL-DB-002	Persist latest Datasheet records to a relational dataset logically named Data .	Must	BR29
DETL-DB-003	Same-day rerun behavior must be idempotent: previously published data for the applicable logical run date/source scope must be replaced transactionally, not duplicated.	Must	BR28

ID	Requirement	Priority	Legacy mapping
DETL-DB-004	Use transactions so a failed database write does not leave an accepted run partially committed.	Must	Reliability NFR
DETL-DB-005	Store run lineage (<code>run_id</code> , source archive checksum/name, processed timestamp) with persisted data or through linked metadata.	Must	Gap filled
DETL-DB-006	Apply database schema changes through version-controlled migrations.	Should	Gap filled

9. Archive and cleanup requirements

ID	Requirement	Priority	Legacy mapping
DETL-ARC-001	Remove extracted temporary files after a successful terminal workflow.	Must	BR6
DETL-ARC-002	Archive/move the original ZIP after successful processing.	Must	BR7
DETL-ARC-003	Retain failed source packages in quarantine or failed-run archive long enough for diagnosis.	Must	Gap filled
DETL-ARC-004	Never delete the only copy of a source package before publication and database commit succeed.	Must	Reliability clarified
DETL-ARC-005	Retention periods for source archives, outputs, logs and quarantine must be configurable/documentated.	Should	Gap filled

10. Exact output model fields

The legacy output schema is preserved:

1. Date
2. Ticker
3. Type
4. Quarter
5. Year
6. Estimated Total Sold
7. Estimated Sold Max
8. Estimated Sold Min
9. Forecast w/o SA Actual
10. Forecast w/o SA Max
11. Forecast w/o SA Min

See [DATA_DICTIONARY.md](#).

11. Datasheet fields

The legacy Datasheet target schema is preserved:

Date, FacilityType, BedSize, Region, Manufacturer, Ticker, Group, Therapy, Anatomy, SubAnatomy, ProductCategory, Quantity, AvgPrice, TotalSpend

12. Non-functional requirements

ID	Area	Requirement
DETL-NFR-001	Reliability	A run must either publish a valid committed result or end in a diagnosable failed/quarantined state; no silent partial success.
DETL-NFR-002	Data integrity	Output/database data must remain consistent with validated source records and same-day rerun policy.
DETL-NFR-003	Robustness	Handle multiple workbooks and multiple model-sheet types without fixed file-count assumptions.
DETL-NFR-004	Performance	Normal end-to-end batch must complete in ≤ 30 minutes in the agreed test environment and dataset size.
DETL-NFR-005	Scalability	Processing design should avoid loading all source workbooks into memory simultaneously when not necessary.
DETL-NFR-006	Security	Encrypt transport, protect secrets, restrict file/database access and avoid logging sensitive workbook contents unnecessarily.
DETL-NFR-007	Observability	Emit run status, duration, files processed, records produced, records rejected, validation failures and terminal error information.
DETL-NFR-008	Maintainability	Workbook schema/anchor rules and business mappings must be centralized and testable.
DETL-NFR-009	Portability	Business logic should not depend on one cloud provider's proprietary runtime unless explicitly approved.
DETL-NFR-010	Recoverability	Operator must be able to rerun a failed batch safely after correcting the underlying issue.

13. CI/CD requirements

- all application, configuration and infrastructure code is version controlled;
- lint/unit tests run automatically;
- representative workbook fixture tests run automatically;
- dependency/secret/security checks appropriate to the implementation run in CI;
- deployable artifact is versioned;
- deployment uses repeatable automation;
- environment secrets are injected at deployment/runtime, not embedded in source.

14. Legacy ambiguity resolutions

BR20-BR22 and BR23-BR25 cell references

The legacy document states that three Empirical values are in column F and three Regression values are in column R, but does not uniquely identify rows/labels. The modern requirement therefore **does not invent row numbers**. The implementation must validate the real workbook format and map values by stable labels/anchors or centralized schema configuration.

BR5 and BR8 same-day filename conflict

The legacy specification requires `model_YYYYMMDD` while also saying every execution creates a new file. Those statements conflict on same-day reruns. The modern rule preserves the canonical business filename while storing run-specific historical artifacts by timestamp/run ID.

FTP/CentOS dependency

The original business outcome was automated cloud file ingestion, not insecure transport as a business goal. Secure SFTP/object-storage transport supersedes plain FTP for the reference implementation. A legacy FTP adapter may be built only as an isolated compatibility component if a trainer explicitly requires it.

15. Complete legacy BR1-BR29 traceability

Legacy ID	Original intent	Modern requirement(s)
BR1	Automate CentOS FTP server on cloud	DETL-ING-001, DETL-ING-002; runtime/transport modernized
BR2	Encrypt <code>/opt</code> storage	DETL-ING-003
BR3	Cron -> Linux script -> unzip -> Python	DETL-ORCH-001, DETL-TR-001
BR4	Create output Excel in <code>/opt/dataout</code>	DETL-OUT-001
BR5	<code>model_YYYYMMDD</code> filename	DETL-OUT-002
BR6	Delete extracted Excel after processing	DETL-ARC-001
BR7	Archive/move ZIP to output area	DETL-ARC-002
BR8	New output per execution	DETL-OUT-003
BR9	Read Estimated Total/Min/Max sold	DETL-TR-007
BR10	Read Forecast Actual/Min/Max	DETL-TR-008
BR11	Multiple model sheets with type	DETL-TR-002, DETL-TR-003
BR12	Read every supported model screen/sheet	DETL-TR-002
BR13	Required output columns	DETL-OUT-004 + Data Dictionary
BR14	Source data becomes output records	DETL-TR-010
BR15	Multiple types produce separate records	DETL-TR-010
BR16	Type from model sheet name	DETL-TR-003
BR17	Missing type -> NULL	DETL-TR-004
BR18	Quarter from Regression Model	DETL-TR-006
BR19	Year from Regression Model	DETL-TR-006
BR20	Estimated Total Sold	DETL-TR-007
BR21	Estimated Max Sold	DETL-TR-007
BR22	Estimated Min Sold	DETL-TR-007
BR23	Forecast w/o SA Actual	DETL-TR-008
BR24	Forecast w/o SA Max	DETL-TR-008
BR25	Forecast w/o SA Min	DETL-TR-008
BR26	Empirical + Regression same type in one record	DETL-TR-005
BR27	Persist output in MySQL <code>AnalystData</code>	DETL-DB-001
BR28	Same-day rerun replaces earlier data	DETL-DB-003

Legacy ID	Original intent	Modern requirement(s)
BR29	Latest Datasheet month persisted to Data with specified columns	DETL-TR-011, DETL-DB-002